



FEDERAL UNIVERSITY OF TECHNOLOGY MINNA

To: The Vice-Chancellor
Federal University of Technology, Minna,
Niger State, Nigeria.

Attention: Director, ITS
Federal University of Technology, Minna,
Niger State, Nigeria.

PROPOSAL FOR INTEGRATING A FINE-TUNED LLM POWERED AI CHATBOT INTO THE FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA OFFICIAL WEBSITE.

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1. Executive Summary:

Imagine having an AI assistant/chatbot trained/fine-tuned with a vast repository of both private and public knowledge related to the school. This knowledge includes the latest articles, updates from the official FUTMINNA website, school rules and regulations, departmental and faculty details, lecturer profiles, and much more. This AI assistant is not only capable of interacting with users but also equipped with the knowledge to assist with a wide range of tasks.

Our proposal aims to develop such an AI chatbot for the Federal University of Technology, Minna (FUTMINNA). The goal is to provide a seamless and intelligent interface for the university community, including both students and prospective students, as well as staff. This AI chatbot will serve as an effortless source of information. Furthermore, it can be seamlessly integrated into the school's website or used as a WhatsApp bot, ensuring accessibility across multiple platforms.

Why We Want to Develop an AI Chatbot for FUTMINNA:

The development of an AI chatbot for FUTMINNA holds immense promise for enhancing the educational and administrative experience within the institution. Our reasons for pursuing this project are as follows:

1. **Real-time Information Access:** The AI chatbot will provide students, staff, and even prospective students with real-time access to essential information, including the latest news, GPA calculation, departmental insights, upcoming events, and more. It will act as a 24/7, ever-ready source of information.
2. **Efficiency and Convenience:** With the AI chatbot in place, the need to sift through vast amounts of data or navigate complex university websites to find relevant information will be eliminated. Users can simply interact with the chatbot, making information retrieval efficient and convenient.
3. **Enhanced User Experience:** The chatbot's ability to provide context-specific responses will significantly improve the user experience. Whether a student is seeking academic

advice, a staff member needs to locate specific administrative data, or a prospective student wants to explore FUTMINNA, the AI chatbot will cater to their needs.

4. **Reduced Workload:** For the university administration, the AI chatbot can assist in automating responses to common queries, freeing up human resources to focus on more complex tasks. This efficiency will result in cost savings and an optimized use of staff resources.

Benefits and Outcomes of the Project:

The development and implementation of the AI chatbot for FUTMINNA will yield a range of benefits and outcomes, including but not limited to:

- **Improved Communication:** Enhanced communication channels will foster a sense of connectedness among students, staff, and the university.
- **Time and Cost Savings:** The AI chatbot will reduce the time and cost associated with manually addressing routine inquiries.
- **Data-Driven Decision-Making:** The chatbot can generate insights from user interactions, aiding in data-driven decision-making for the university.
- **Competitive Edge:** FUTMINNA will gain a competitive edge by embracing innovative technology to meet the evolving needs of its community.
- **Scalability and Customization:** The AI chatbot can be scaled and customized to meet future needs, making it a valuable long-term investment.

By developing this AI chatbot, we aim to create a more accessible, efficient, and user-friendly environment for FUTMINNA's academic and administrative activities, reinforcing the institution's commitment to excellence.

2. Introduction:

Greetings! I am Shedrack Oluwamo (Full-Stack Developer) a 200-Level Project Management student at FUTMINNA, with over 7 years of experience in software development. My professional journey has led me to develop AI-driven compliance solutions and anti-money laundering systems that are currently in use by esteemed financial institutions, including Union Bank, KeyStone Bank, Piggy Vest, and Trust Bank Gambia. As a testament to my commitment to excellence, I currently serve as a team leader within the IT department at Adroit Solutions, where I have harnessed my comprehensive skills and deep expertise to contribute significantly to the creation of Pye.

Purpose and Significance of the Proposal:

The purpose of this proposal is to introduce an initiative that has the potential to revolutionize the way information is accessed and disseminated within FUTMINNA. We seek to develop an AI chatbot powered by advanced language models such as Llama-2 ChatGPT and Bard, employing our Pye.Ai platform. This chatbot, with its remarkable capabilities, will serve as a go-to resource for students, non-students, and staff, offering a seamless and efficient means of obtaining information. Our proposal is significant because it represents a transformative step toward modernizing communication and providing a centralized platform for information retrieval.

This chatbot will not only enhance the accessibility of data but will also offer unprecedented customization, allowing it to adapt to the unique needs and preferences of FUTMINNA's diverse user base. By integrating various data sources, such as website pages, documents, and databases, we aim to create a chatbot that is both intelligent and versatile, thus improving the overall experience of our school community.

We believe that this project aligns with the forward-thinking vision of FUTMINNA and will contribute to a more connected and informed campus. We are excited about the opportunity to collaborate with the school authorities to bring this vision to life, and we are confident that the introduction of this AI chatbot will have a lasting and positive impact on our institution.

3. Project Overview:

Pye.ai is an advanced AI-powered chatbot building platform. It harnesses the fine-tuning of existing Large Language Models (LLM), including both publicly available LLMs such as Mini-GPT and Llama-2, as well as private LLMs like ChatGPT, Bard, and Anthropy. With Pye.Ai, there's no need to train a model from scratch; you can create highly customizable chatbots with minimal data. These chatbots are proficient in natural language understanding and generation, making them suitable for various applications, including customer support, information retrieval, and task automation.

The chatbot created using Pye.Ai can be seamlessly integrated into the School official website, ensuring easy access for students and staff. Users will be able to initiate conversations with the chatbot by clicking on a designated chat interface. Once engaged, the chatbot can engage in dynamic conversations, offering information, assistance, and guidance as needed.

To guarantee the chatbot's accuracy and relevance, we have implemented a robust data integration system (ETL) and Prompt Builder. This system is designed for real-time information retrieval from various school resources, including website pages, documents, databases, and other data repositories. Leveraging Application Programming Interfaces (APIs), web scraping, and structured data retrieval methods with a custom ETL Pipeline integrated into Pye, the chatbot will have real-time access to the most up-to-date information. This integration will ensure that users receive the most current and accurate responses.

We understand the importance of tailoring the chatbot's responses to meet the specific needs of FUTMINNA. To achieve this, Pye.Ai provides a user-friendly prompt builder interface. Users, administrators, or content providers can create custom prompts and responses, defining the chatbot's behavior for various scenarios. For example, it allows you to create prompts/chatbots for frequently asked questions, specific departmental inquiries, or event announcements. Additionally, our platform enables dynamic customization, ensuring that the chatbot can adapt to evolving requirements, thereby enhancing its utility and relevance.

4. Benefits:

Benefits for Students:

1. **24/7 Accessibility:** Students can access information and support at any time, including outside regular office hours.
2. **Instant Answers:** Immediate responses to common queries, such as class schedules, exam dates, and registration information.
3. **Personalized Guidance:** Chatbots can provide personalized academic advice, recommend relevant courses, and suggest resources based on a student's academic history.
4. **Reduction of Administrative Burden:** Students can quickly find information without relying on administrative staff, freeing up their time for more complex tasks.
5. **Onboarding and Orientation:** New students can receive instant help with orientation, campus navigation, and general inquiries.

Benefits for Non-Students (Parents, Prospective Students, Alumni, etc.):

1. **Admissions and Enrollment Information:** Prospective students and parents can get information about admission criteria, application deadlines, and enrollment processes.
2. **Alumni Engagement:** Alumni can access information on events, reunions, and stay connected with the institution.
3. **Community Engagement:** The chatbot can provide information about public events, conferences, and outreach programs hosted by the school.
4. **Donations and Fundraising:** A chatbot can guide interested parties on how to donate, support scholarships, or contribute to the school's initiatives.

Benefits for Staff:

1. **Reduced Workload:** Administrative staff can offload routine inquiries to the chatbot, allowing them to focus on more complex tasks.
2. **Efficiency:** Staff can quickly access information, documents, and resources by using the chatbot, streamlining their workflow.
3. **Internal Resources:** The chatbot can provide staff with links to internal documents, policies, and guidelines.
4. **Training and Support:** Staff can use the chatbot for on-the-job training and immediate troubleshooting.
5. **Feedback and Improvement:** The chatbot can gather feedback from users, helping staff identify areas for improvement in various services.

5. Scope of Work:

Tasks and Responsibilities of our Team

- **Project Manager (Oluwamo Shedrack Tunde):** Responsible for overall project coordination, communication with school authorities, and ensuring project milestones are met.
- **AI Chatbot & Web Developers (Oluwamo Shedrack , Akano Ismail, Adamu Farouq):** Responsible for developing and configuring the Pye.Ai chatbot, including customizing its behavior and integrating it with the school website.
- **Data Integration Specialists (Akano Ismail):** Responsible for data extraction pipelines.
- **Quality Assurance Team:** Responsible for testing the chatbot to ensure it functions correctly and meets the specified criteria.
- **User Training and Support (Oluwamo Shedrack Tunde):** Responsible for providing training and support to the School IT Team regarding the platform usage, if needed.
- **Funding:** While our team is committed to the project, some costs are associated with server maintenance, LLM fee, and we request a budget allocation to cover these expenses.

Resources or Support needed from the School:

- **Access to a Dedicated Server Space:** We require secure and dedicated server space for hosting the Pye.Ai platform to ensure data privacy. This is essential for safeguarding private user data, including website pages, documents, and databases, remains confidential and protected. This server will be exclusively used for hosting the Pye.Ai chatbot, and strict security measures will be implemented to prevent unauthorized access.
- **Access to School Databases:** We will need read-only access to specific school databases, such as the course catalog and event calendars, to ensure the chatbot can provide accurate and up-to-date information.

- **Technical Support:** We kindly request that the school's IT department provide technical support and assistance during the integration phase to ensure a smooth and secure deployment.
- **Data Privacy and Security Compliance:** We will need the school's cooperation in ensuring that all data integration and usage align with the school's data privacy and security policies.
- **Testing Environment:** It would be highly beneficial to have access to a staging or testing environment on the school's web server to conduct thorough testing of the chatbot's functionality before it goes live.

6. Technical Requirements for Integrating the Chatbot:

1. Hosting and Infrastructure:

Pye.Ai requires a dedicated school server (recommended) or a cloud-based hosting environment (optional) with a minimum of 8GB of RAM and 2 vCPUs to ensure optimal performance.

2. Data Integration:

The platform is already included with a custom ETL service to fetch information from the school's databases, website pages, and documents stored in supported formats (PDFs, Word documents, etc.).

3. Data Privacy and Security:

Data privacy is of utmost importance. Pye.Ai will implement end-to-end encryption for all data transmissions. Access to private data will be restricted, and the system will comply with relevant data protection laws.

4. Interaction with FUTMINNA Website:

The chatbot will be integrated into the FUTMINNA website as a user-friendly widget, accessible from the bottom right corner of every page. Users can initiate conversations by clicking on the chatbot icon.

5. User Feedback and Learning:

The chatbot automatically trained itself from user interactions and improve its responses. User feedback mechanisms will be implemented to collect suggestions and correct inaccuracies.

6. Compatibility:

The chatbot is designed to be responsive and compatible with major web browsers and devices, ensuring a consistent and user-friendly experience for all users.

7. Project Timelines:

Day	Milestone
Day 1	Project kickoff and team orientation
Day 2 to 7	Requirement gathering and analysis
Day 8	Technical Setup and Installation
Day 9 to 11	Data integration setup
Day 12	Chatbot creation & customization
Day 13 to 14	Prompt Creating and Testing with Prompt Builder
Day 15	Initial testing and debugging
Day 16	Beta testing with a select group of users
Day 17 to 20	Feedback analysis and refinements
Day 21	Privacy and Security Audit
Day 22	Integration with official website
Day 23 to 24	User training
Day 25	Full Launch
Day 25 to 27	Monitoring and Evaluation
Day 28	Documentation & Reporting
Day 29	Project Closure and Handover

8. Budgets:

Please note that all essential services required for the Pye.Ai chatbot project, including data integration, customization, and other functionalities, are fully built into the Pye core platform. This design choice eliminates the need for external services, ensuring that the project's expenses are limited to the monthly fee for the Language Model (LLM), making the overall cost effectively zero beyond this subscription fee.

Item	Cost per Month (Naira)	Duration	Total Cost (Naira)
LLM Monthtly Fee (ChatGpt)	16000	6 months	96000

9. Team Members:

Our project team consists of three dedicated and passionate individuals.

OLUWAMO TUNDE SHEDRACK (TEAM LEADER)

Oluwamo is a 200-Level Project Management student with a strong background in project implementation (Application Deployment) and experience in LLM, AI-driven solutions and Full-Stack Development. He will oversee project coordination, client interactions, and technical development.



AKANO ISMAIL (DATA ANALYSTS)

Ismail, a 200-Level Agricultural Extension Student with over 3 years of experience specializing in data analysis, web development and scraping technologies. Ismail will play a crucial role in ensuring the integration of required data and other related information into the bot.

ADAMU FAROUQ (WEB DEVELOPER AND TESTING EXPERT)

Adamu, a 200-Level Quantity Survey student, brings a strong technical background in web development and expertise in software testing. He will be responsible for the technical development and testing phases of the chatbot.

10. Approval Letter:

Dear Sir/Ma,

We, the undersigned Oluwamo Shedrack Tunde and Akano Ismail, on behalf of the Pye.Ai team, hereby request formal approval to integrate our AI chatbot platform into the FUTMINNA official website. Our project aims to enhance user experience and accessibility to information, benefiting students, non-students, and staff members. We believe that this integration will be a valuable addition to the school's digital ecosystem.

We kindly request your consideration and support for this project, and we are prepared to provide further details, conduct demonstrations, and address any concerns you may have.

Thank you for your time and attention to this proposal. We look forward to your positive response.

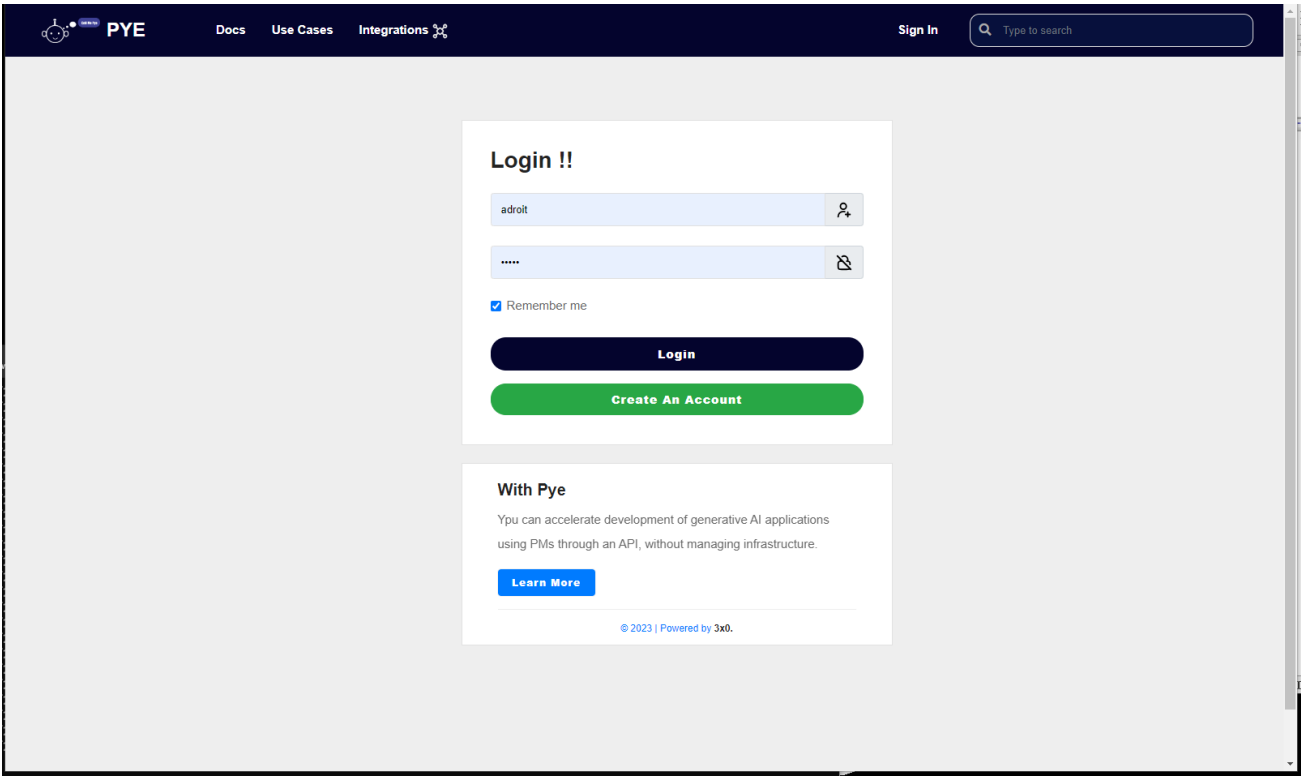
Sincerely,

Oluwamo Shedrack Tunde

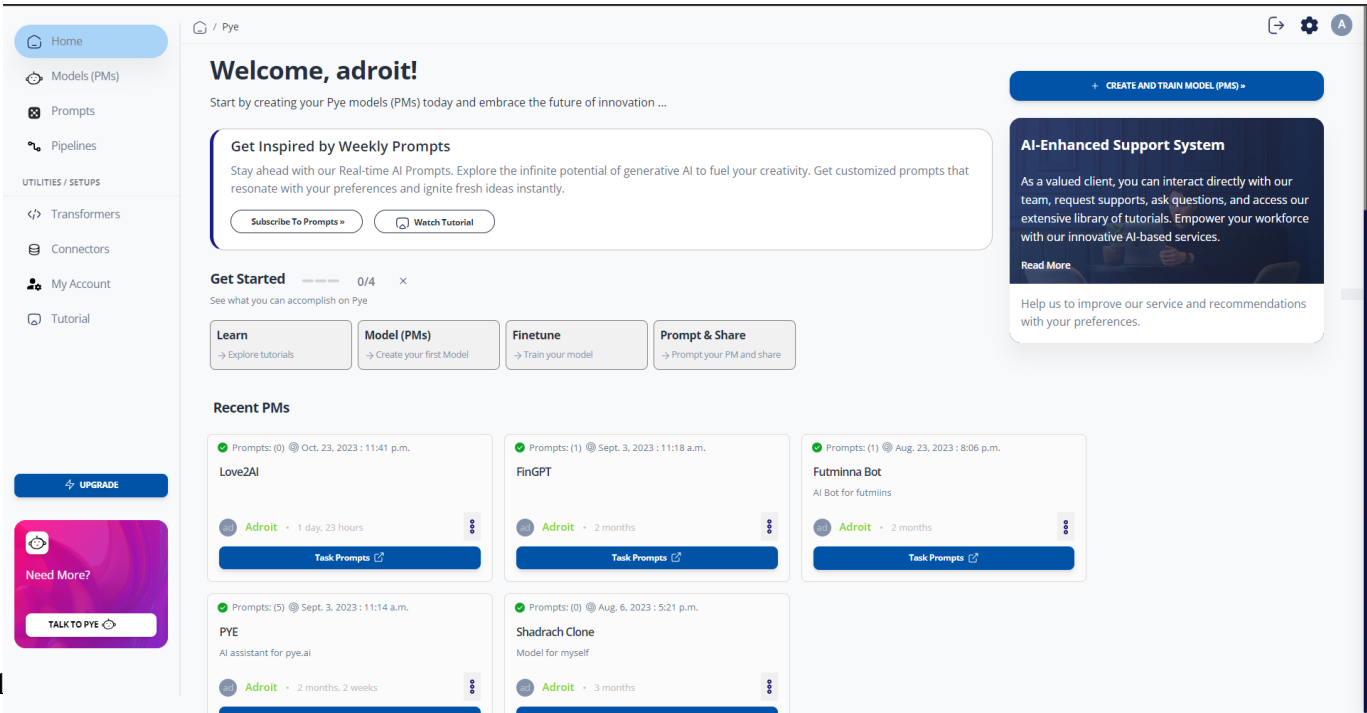
09158929344

12. Pye.AI Screenshots:

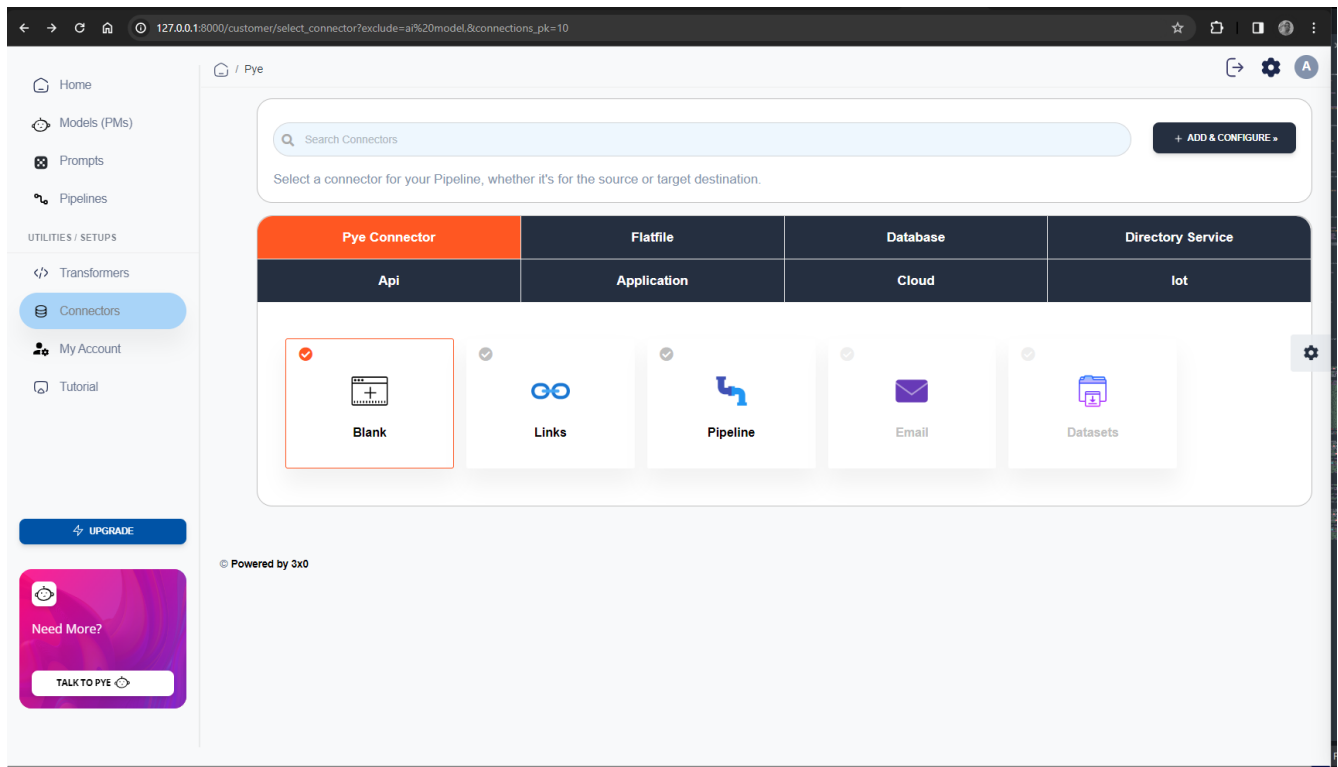
Pye.AI - Login Page



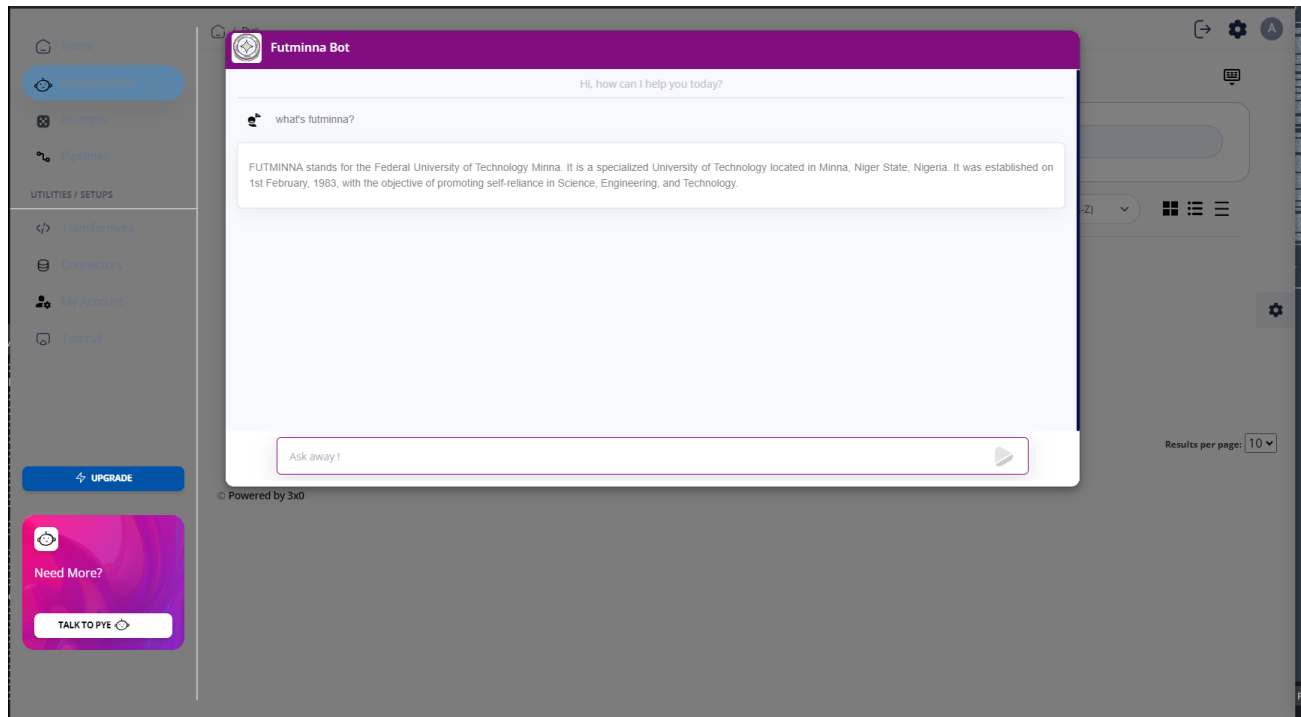
Pye.AI - Home Page



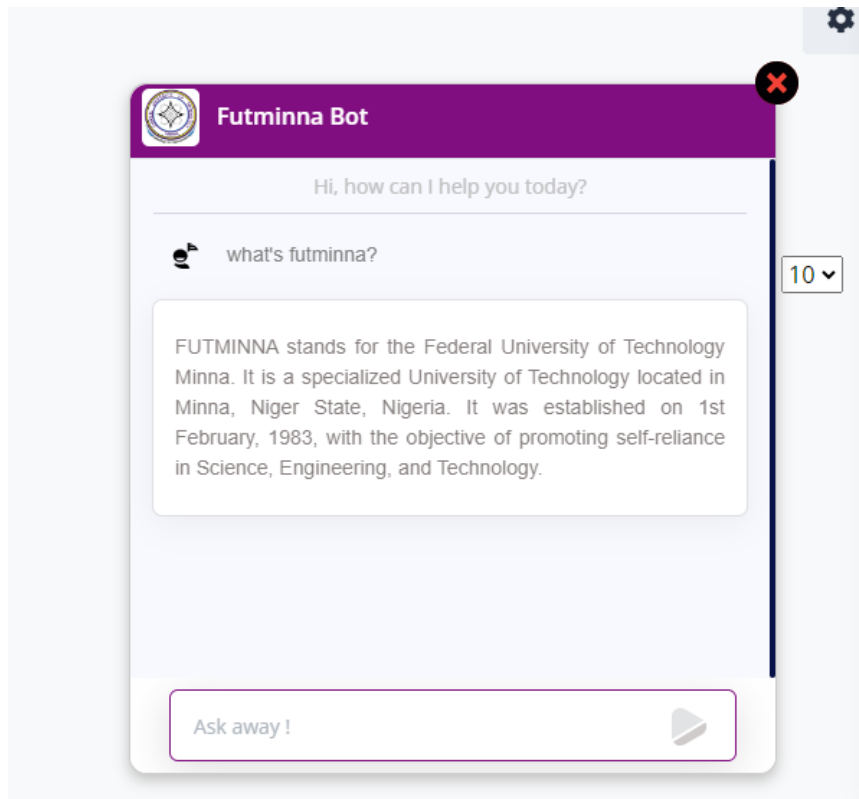
ETL / Data Integration System



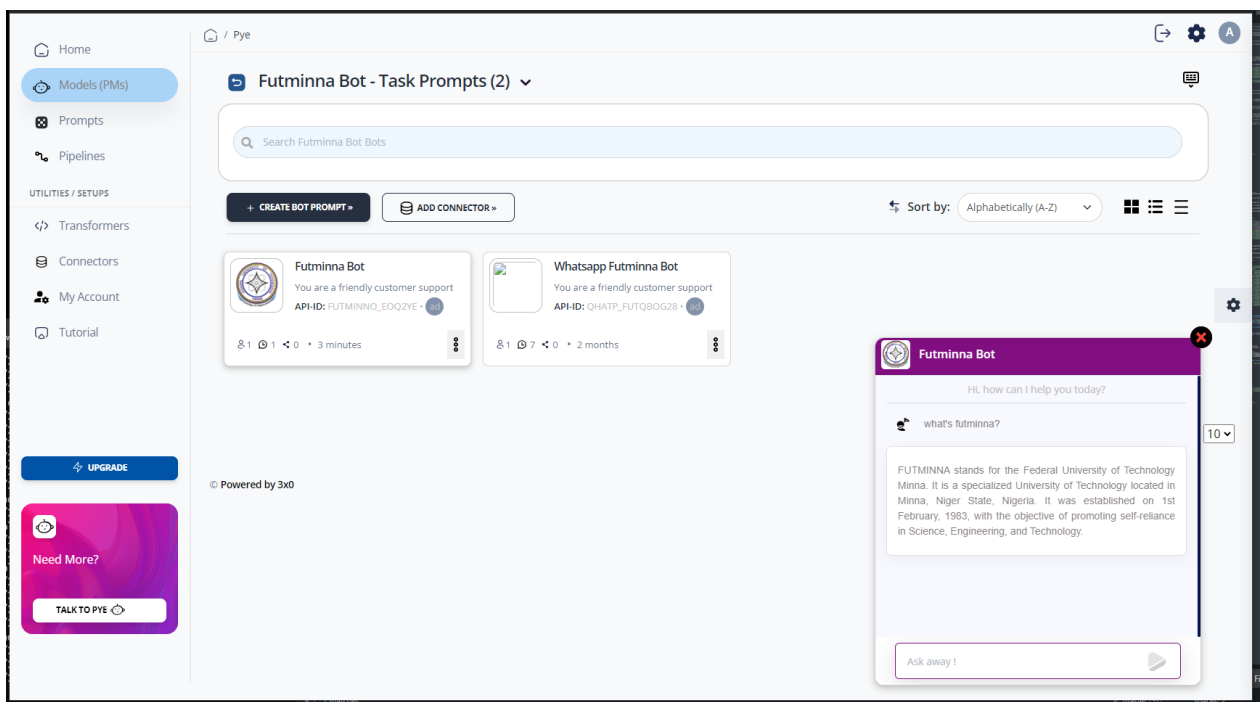
ChatBot Test Preview



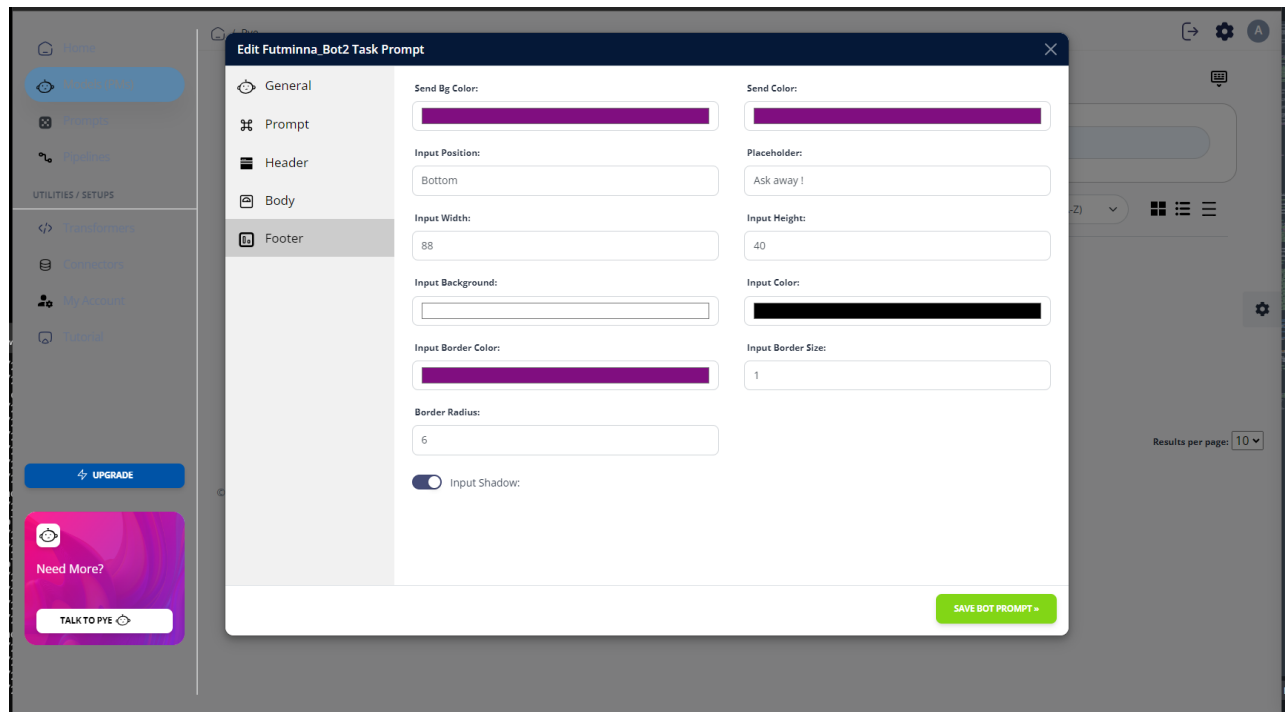
Integrated Chatbot Sample



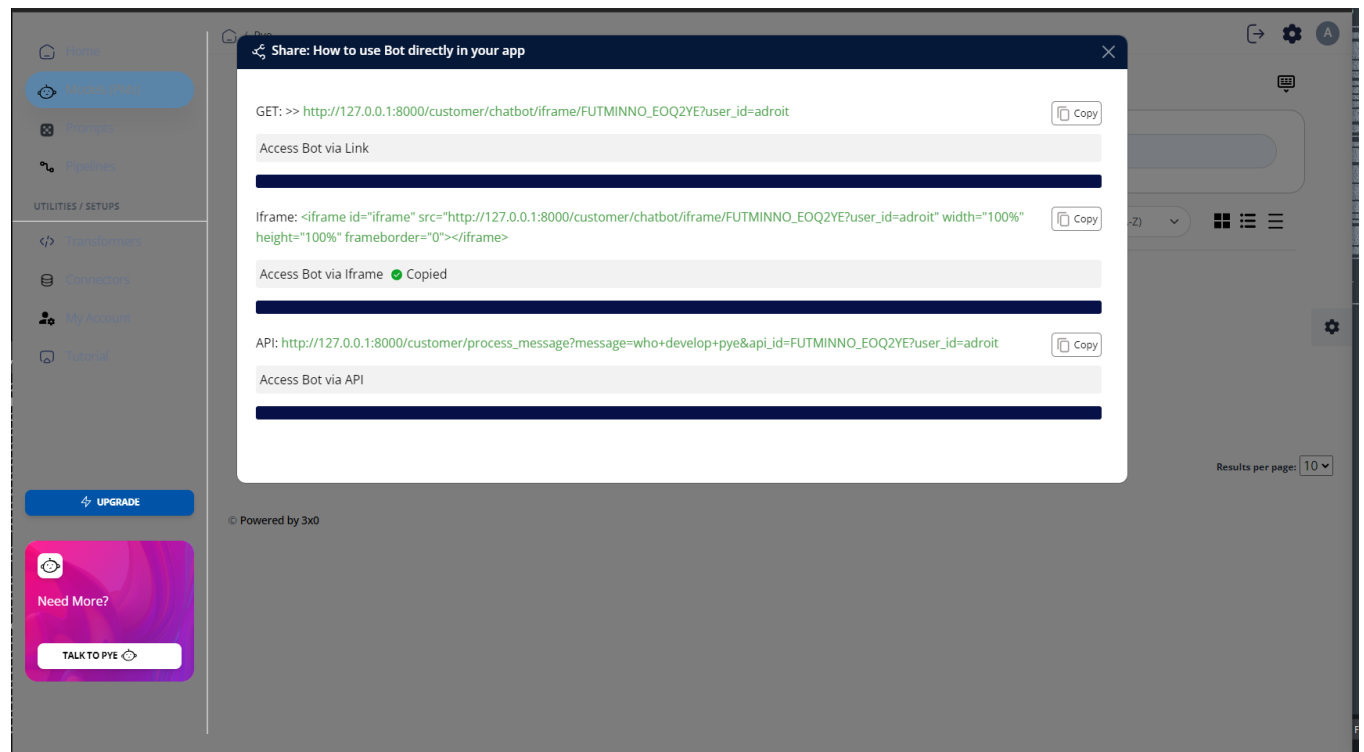
Sample Chatbot Integration on Pye



Chatbot Customization Builder



ChatBot Share (API / Iframe)



11. Conclusion:

In summary, this proposal outlines our intention to integrate Pye.Ai, an AI chatbot platform developed by our team, into the FUTMINNA official website. The proposal elucidates the technical and practical aspects of this integration, covering the objectives, scope, technical requirements, privacy considerations, budget, and the timeline for implementation.

Furthermore, it highlights the advantages that this endeavor promises for our academic institution, its students, staff, and visitors.

The integration of the Pye.Ai chatbot into the FUTMINNA official website offers a multitude of benefits. This innovative addition will provide a user-friendly interface for students, non-students, and staff to easily access information and receive prompt assistance. By harnessing the power of AI, we empower our institution with a customizable chatbot capable of efficiently and accurately answering queries, navigating databases, and providing relevant information. The chatbot will enhance the overall user experience, improving accessibility and reducing response times. Moreover, it offers a cost-effective solution that aligns with FUTMINNA's commitment to technological innovation and efficient communication.

This technology will not only streamline administrative processes but also create a welcoming and interactive environment for both current and prospective members of our academic community. The chatbot integration has the potential to transform the way we engage with our stakeholders, ultimately enhancing the school's reputation and accessibility in the digital age.

12. Contact Information:

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